

	Cassandra	MongoDB	HBase	Neo4j	Riak
Developed using	Java	C++	Java	Java	Erlang and C
Database type	Wide-column stores	Document stores	Big Table stores/ Document stores	Graph databases	Wide-column stores
USP (Unique selling point)	Almost SQL-like. Handles massive datasets	Almost SQL-like. Contains support for querying and indexing	Massively scalable, up to billions of rows by millions of columns	Used for complex, interconnected data processing	Capable of storing BLOBs
Protocol	CQL3 and Thrift	BSON	HTTP and Thrift	HTTP and REST	HTTP and REST
License	Apache	AGPL	Apache	GPL	Apache
Usage	Data collection, logging data analysis, Web analytics data processing	Integrates well with Web based applications	Search engines, logging data analysis	Used for searching and plotting map data, network topologies and social analytics	Storing large objects, files

Table 1: A comparison of commonly used NoSQL databases

- **Easy and quick scaling:** NoSQL databases can scale out much faster than traditional RDBMSs because of the monolithic architecture followed by most of the latter variety. Scaling can be achieved by simply adding more nodes to the NoSQL database cluster according to the data storage needs.
- **Schema-less data storage:** NoSQL databases can store and handle all types of data, whether they are structured, non-structured or even semi-structured. This is possible as NoSQL databases have no fixed data model. This flexibility enables organisations to access and store much larger quantities of data, as and when required.
- **Quick and easy data manipulation:** While traditional relational databases use their own specific query languages to perform any data manipulations and operations, NoSQL databases use JSON and other such object-oriented APIs to perform the same tasks. This makes it easier for developers to code and develop their applications and integrate them to NoSQL databases.
- **No SPoF (Single Point of Failure):** The best part about using a NoSQL database is that most of the underlying complex tasks such as high availability with automatic load balancing, clustering, etc, are all performed and managed by the database itself. Most of the current and popular NoSQL databases come with inbuilt fault tolerance and can continue to function even if a couple of nodes go down. NoSQL databases have a variety of unique use cases and deployment scenarios. Some of the most common use cases of these databases are:
 - **Data mining of logs:** NoSQL databases can be used to store and analyse large datasets of logs. A good example of one such NoSQL database is Cassandra, which can manage and mine data out of log files that are extremely large in size.

- **Real-time analytics:** NoSQL databases are very popular when it comes to mining data that is constantly changing. Organisations can track the real-time performance of their systems, monitor stocks or track unique user hits on a website when a NoSQL database is used.
- **Big Data processing:** Perhaps one of the most interesting and key uses of NoSQL databases is in the Big Data industry, where a NoSQL database can be used to store, analyse and provide useful insights on data that is simply too massive for a traditional RDBMS to handle.

Introducing Apache CouchDB

CouchDB is a highly concurrent NoSQL database that is designed to scale horizontally across numerous devices with ease, and be fault-tolerant and reliable. It is a document-oriented NoSQL database that uses many programming languages such as JSON to store data, JavaScript to query the data using MapReduce, and HTTP as an application interface.

Unlike traditional RDBMSs, CouchDB stores its databases as a collection of independent documents. Each of these documents contains its own schema and metadata. The metadata contains additional attributes such as the revision number and replication information, which can be used by developers during the replication process to synchronise two or more CouchDB databases.

Installation

CouchDB can be installed both via packages as well as from source. Installation from packages is quite simple in case you are using a Debian or Ubuntu system. All you need to do is run the following command at the terminal:

```
# sudo apt-get install couchdb
```